

Analytical Report on Global Diabetes Data



Date: December 6th, 2024

By Soohan Woo (A20514858)

Table of Contents

Abstract.....	3
Introduction.....	3
Literature Review.....	4
Methodologies.....	10
Results.....	16
Conclusion.....	31
Project Breakdown.....	33
References.....	34
Appendix.....	35

Abstract

Diabetes is one of the most prevalent chronic diseases in the world, affecting hundreds of millions of people around the world and serving as an increasing cause of concern for medical experts. As part of the Business Analytics and Data Science program at Oklahoma State University, this study conducted in the course MSIS 5193 aims to investigate worldwide data on diabetes and identify risk factors associated with diabetes, joining in a large movement dedicated to predicting and preventing diabetes. This study involved extracting data from published diabetes data, with an emphasis on how BMI (Body Mass Index), age, and gender affect one's risk of developing diabetes. Through the employment of multiple analytical techniques based on the Python programming language, this study prepared the selected data, connected multiple datasets from different people groups, and assessed the relationship between the more commonly known risk factors and the risk of developing diabetes. These results and conclusions are hopefully beneficial to the global effort to develop increasingly accurate predictive models regarding diabetes development.

Introduction

Diabetes Overview

Throughout the world, diabetes is a serious and potentially life-threatening disease that affects hundreds of millions of people today. It is a chronic disease that hinders the body from producing enough insulin to regulate blood sugar levels, leading to blood sugar levels that high enough that they could even cause death. Due to the widespread nature of this disease and the urgency to counter its spread, much data have been produced and published publicly that associate a long list of features, such as BMI, age, gender, smoking habits, and dietary habits with the presence of diabetes in an individual. Much research has gone into developing predictive models for diabetes, and the information from the selected datasets will help us to draw holistic conclusions about the effects of certain features on diabetes.

Objectives

This study aims to complete several connected objectives. Firstly, we will thoroughly prepare the available data, using a dataset that includes all of our features of interest and possesses a sample size large enough to generate meaningful analysis. It is important that the data are representative of humanity in general, instead of only representing a specific, localized people group. Therefore, we have chosen a dataset explicitly generated for the purpose of diabetes prediction. This dataset was generated primarily from Electronic Health Records (EHRs), which are digital versions of patient health records and include all of our features of interest. This dataset contains a large amount of patient data, and is formatted in a way to help develop machine learning models. Afterwards, we will assess whether or not there is a significant relationship between the risk factors (BMI, age, and gender) and the presence of diabetes. The appropriate type of analysis will be done with each risk factor, since the risk factors differ from each other when it comes to the type of data that they consist of. Various statistical tests and association rule mining will be applied to the relationship between the risk factors and the presence of diabetes in order to draw a conclusion on whether or not these risk factors significantly influence the risk of developing diabetes. By default, our hypothesis is that all our chosen risk factors are significant predictors of the presence of diabetes. In this analytical process, we will generate models that can be useful for predicting diabetes based on significant risk factors. These models will allow us to test individual patient records and predict whether or not patients that currently do not have diabetes are at risk of diabetes, hopefully contributing to the global movement to develop preventive measures for diabetes.

Literature Review

Chaudhuri I., Alam M.S., Kamrujjaman M. “Novel Risk Factors for Diabetes: A Comprehensive Analysis for Enhanced Disease Diagnosis.” *International Journal of Ground Sediment & Water*, 19, 1355–1382 (2024). <https://doi.org/10.5281/zenodo.10791178>.

This study uses diabetes data collected from the Iraqi community to evaluate diabetes-related biomarkers, and to create a more precise diagnostic framework for diabetes (Chaudhuri,

2024). The associated dataset contains medical records and laboratory findings from 1,000 people who had diabetes, did not have diabetes, or were expected to develop diabetes. The dataset includes factors such as blood sugar level, age, gender, and BMI. The patients are categorized as diabetic, non-diabetic, or predicted diabetic.

Initially, this study prepares the data by carefully selecting characteristics that are relevant to diabetes, such as BMI and cholesterol level, and by removing data points identified as outliers in order to avoid skewing the analytical results. The study uses strip plot graphs to visualize the relationships between each characteristic and diabetes, drawing conclusions on the relationships based on the generated statistics. The selected biomarkers are then used to conduct correlation analysis with the presence of diabetes, and conclusions are drawn about how and to what degree each variable is correlated with the risk of diabetes. The study concludes that, amongst other factors, BMI and age are likely significant risk factors for diabetes.

Chen Y., Zhang X.P., Yuan J., Cai B., Wang X.L., Wu X.L., Zhang Y.H., Zhang X.Y., Yin T., Zhu X.H., Gu Y.J., Cui S.W., Lu Z.Q., Li X.Y. “Association of body mass index and age with incident diabetes in Chinese adults: a population-based cohort study.” *BMJ Open* 8(9):e021768 (2018). doi:10.1136/bmjopen-2018-021768.

This study aims to identify and explore an association between obesity, age, and diabetes (Chen, 2018). The associated dataset contains a record of 211,833 Chinese adults over 20 years old across 32 sites and 11 cities in China, all of whom did not initially have diabetes, and contains data on their age and BMI levels. The individuals are studied across multiple visits, to identify the influence of BMI and age on those who ended up developing diabetes.

The study expands across a time frame of up to several years between 2010 and 2016 as the researchers follow up with the individuals. The SAS software is used to perform statistical analysis and develop confidence intervals regarding the association of the risk factors and the risk of incident diabetes. By drawing linear associations between the risk factors and the risk of incident diabetes, the study finds that BMI is independently associated with a higher risk of developing diabetes, although it also found that the effects of BMI on incident diabetes were more pronounced in younger adults. Additional factors, such as smoking habits and family history of diabetes, were also considered in relation to the risk of incident diabetes.

Kapur A., Seshiah V. “Women & diabetes: Our right to a healthy future.” *Indian J Med Res* 146(5):553-556 (2017). doi: 10.4103/ijmr.IJMR_1695_17.

This study makes a case that women are more severely impacted by diabetes than men are (Kapur, 2017). By drawing from pre-existing studies and research results, this paper lists the consequences of diabetes, such as an increased risk of heart attack, and cites relevant data to show that the consequences are more severe for women with diabetes than for men with diabetes.

The study then examines the occurrence of hyperglycemia in pregnancy. Risk factors such as BMI, socio-economic status, and body weight, are linked to the risk of diabetes during pregnancy using existing results from past correlation analyses. The study draws conclusions on how individuals born in environments with considerable levels of undernutrition are at an increased risk of diabetes during pregnancy, and makes suggestions for changing policies and advancing greater scrutiny in healthcare to prevent transgenerational risk transmission.

Smith J.W., Everhart J.E., Dickson W.C., Knowler W.C., Johannes R.S. “Using the ADAP Learning Algorithm to Forecast the Onset of Diabetes Mellitus.” *Proc Annu Symp Comput Appl Med Care* 261–5 (1988).

This study aims to develop a neural network model by using it to predict the onset of diabetes in a high-risk population of Pima Indians (Smith, 1988). This dataset has been widely used in the study of diabetes, and hundreds of medical and laboratory results have been recorded from the population by the National Institute of Diabetes and Digestive and Kidney Diseases since 1965. The dataset includes characteristics associated with diabetes, such as glucose levels, BMI, and age, and records the presence of diabetes as a simple binary variable.

The study begins by selecting for certain individuals to use the data of based on certain criteria, and ends up using data from female subjects over 20 years of age. From the resulting 768 examinations, the study uses an early neural network model, ADAP, to forecast the onset of diabetes based on eight chosen features, including BMI and age. The accuracy of the model is tested using specificity and sensitivity values, as well as ROC curves, and conclusions are drawn that the model, although still under development, fits the data well enough to be useful for

predicting diabetes. No definitive conclusions are drawn about which variables significantly increase the risk of diabetes, although preliminary results are provided that indicate that certain variables may be more associated with the onset of diabetes than others.

Strack B., DeShazo J.P., Gennings C., Olmo J.L., Ventura S., Cios K.J., Clore J.N. “Impact of HbA1c measurement on hospital readmission rates: analysis of 70,000 clinical database patient records.” Biomed Res Int. (2014). doi: 10.1155/2014/781670.

This study aims to draw an association between HbA1c, a blood test that measures blood sugar levels, the presence of diabetes, and the early readmission of a patient (Strack, 2014). The dataset draws almost 70,000 inpatient diabetes encounters from a large clinical database containing data on 74 million unique encounters corresponding to 17 million patients. The dataset includes data on biomarkers such as weight, gender, and age, as well as information on the number of times each patient was readmitted. The dataset also records laboratory results, such as from the HbA1c tests, and whether or not each patient ended up developing diabetes.

The study begins by collecting the large amount of data that it will use, and by removing features that had a high percentage of missing values, such as weight, which was missing 97% of its values. The researchers then use multivariable logistic regression to fit a relationship between the measurement of HbA1c and early readmission while controlling for variables such as the type of admission and demographics. The employment of logistic regression shows the value of this statistical method in assessing the relationships between variables at an academic level. The researchers draw the conclusion that the measurement of HbA1c for patients with diabetes is a useful predictor of readmission rates, leading to a suggestion that, if possible, HbA1c should be tested for more frequently because of the valuable insight it provides into indicating the presence of diabetes.

Tigga N.P., Garg S. “Prediction of Type 2 Diabetes using Machine Learning Classification Methods.” Procedia Computer Science, Volume 167, Pages 706-716, ISSN 1877-0509 (2020). <https://doi.org/10.1016/j.procs.2020.03.336>.

This study aims to analyze the risk of diabetes based on lifestyle and family background (Tigga, 2020). The associated dataset consists of 952 instances of using an online and offline questionnaire to ask 18 questions about health, lifestyle, and family background. Different machine learning algorithms are used to predict the risk of Type 2 diabetes, and the fully trained model is tested on both the collected data and the popular Pima Indian Diabetes database.

The study begins by research on diabetes itself, to better understand what causes the disease. After acquiring data from the general public through questionnaires, the data are cleaned and transformed appropriately. Various analytical methods are applied to the data in order to identify which method develops a model that most accurately predicts diabetes based on relevant characteristics. Several machine learning classification techniques are employed to develop the model, such as logistic regression, the k-nearest neighbor method, and the random forest classifier. Upon testing the accuracy of the different models against each other, the study finds that the random forest classifier produces the highest accuracy for both datasets, and also notes that age is one of variables with the highest significance in predicting diabetes.

Wu Y., Ding Y., Tanaka Y., Zhang W. “Risk factors contributing to type 2 diabetes and recent advances in the treatment and prevention.” *Int J Med Sci.*;11(11):1185-200 (2014). doi:10.7150/ijms.10001.

This study seeks to identify genes, lifestyle choices, and other factors that have contributed to a rapid increase in the incidence of Type 2 diabetes (Wu, 2014). The associated dataset consists of recent statistical data recording the presence of diabetes and pre-diabetes in people older than 20 years old from several different countries. Data on relevant factors, such as blood glucose levels and physical inactivity, are also available in the dataset.

This study begins by testing for correlation between different genes and the onset of Type 2 diabetes. The correlation analysis controls for factors such as the individual’s nationality. Then, the study tests for correlation between lifestyle factors, such as smoking and alcohol consumption, and the onset of Type 2 diabetes. The application of correlation analysis continues in studying the relationship that the gut metagenome and different vitamins have with controlling for or increasing the risk of diabetes. The study then transitions into a more informational phase,

where it lists complications with diabetes, such as cardiovascular disease, and provides the commonly used diabetes treatments.

The study concludes with a comparison of different prediction models, such as ones that use noninvasive measures like BMI and age, and ones that use genetic markers. The study lists the information that is required to develop each unique prediction model, and emphasizes the need for more prediction strategies to develop more effective intervention trials and prevention measures that would help avoid the occurrence of diabetes.

Xie Z., Nikolayeva O., Luo J., Li D. “Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques.” *Prev Chronic Dis* 16:190109 (2019).

<http://dx.doi.org/10.5888/pcd16.190109>

This study aims to develop predictive models to identify risk factors for Type 2 diabetes, with a focus on the population of the United States (Xie, 2019). The associated dataset consists of cross-sectional data on 138,146 participants, including 20,467 with Type 2 diabetes, from the Behavioral Risk Factor Surveillance System. The data contain records of factors associated with diabetes, such as age and BMI, as well as the presence of diabetes in each of the participants.

The study begins by selecting for data points based on certain criteria, eliminating participants younger than 30 years old and also participants who had diabetes while pregnant, in order to control for variables deemed to skew the analytical results. Variables with missing values were also removed from the analysis. Afterwards, eight machine learnings were built using different techniques, such as support vector machine, decision tree, logistic regression, random forest, and neural network. The researchers test the accuracy of the different models using sensitivity, specificity, and AUC values, and conclude that the neural network model gave the best model performance while the decision tree model has the highest sensitivity, and thus the highest detection rate. The models are used to confirm the significance of more well-known risk factors, such as BMI and age, while two new potential risk factors are identified as well: sleeping time and frequency of checkup. It is also noted at the end of the study that logistic regression is a popular technique for developing predictive models for Type 2 diabetes.

Methodologies

The dataset consists of 100,000 data points and 9 variables, with each data point representing data collected from a unique patient, provided by multiple healthcare providers. Of the 9 columns available, we selected 4 that we consider relevant to our analysis' objectives: 'gender' (representing the gender of the patient), 'age' (measuring the age of the patient), 'bmi' (measuring the BMI of the patient), and 'diabetes' (recording 0 for patients without diabetes and 1 for patients with diabetes). In order to complete a holistic analysis, we have organized our methodologies into several steps:

1. Data Understanding
2. Data Preparation and Feature Engineering
3. Association Rule Mining
4. Logistic Regression
5. Testing Predictive Models

Data Understanding

We began our analysis by simply exploring the subset of the data that we have chosen for our objectives. Using the describe function from the pandas library in Python, we extracted key statistics, such as the count and mean, of each of our variables. Using the matplotlib.pyplot and seaborn libraries in Python, which are both powerful libraries containing several data visualization tools, we visualized key aspects of the variables. Histograms were used to display the frequency distributions of our variables, boxplots were used to visualize the relationships between our continuous independent variables and our target variable of the presence of diabetes, and a bar chart using the countplot() function was used to visualize the relationship between our binary independent variable and our target variable.

Each type of visualization served an essential purpose before applying our analytical tests. The histograms gave us insight into the distributions of our variables, so that we could consider factors such as skewness in our analysis. The boxplots and bar chart gave us a preliminary glimpse into the relationship between our risk factors and our target variable, so that

we could have an initial understanding of the interplay between the variables before conducting more specific analysis. The key statistics that were generated also provided insight into each individual variable, allowing us to better comprehend the distribution of the variables and to potentially expect outliers. This preliminary exploration of the data allowed us to have a general understanding of how and to what degree each independent variable affected the presence of diabetes.

Data Preparation and Feature Engineering

After having a general understanding of the data, we conducted several actions to prepare the data for analysis, such as checking for and removing rows with missing values, converting variables into categorical variables if required for analysis, and creating dummy variables as necessary. For the purposes of our analysis, we extracted a subset of our data containing our variables of interest to perform our data preparation and feature engineering steps on: 'gender' (representing the gender of the patient), 'age' (measuring the age of the patient), 'bmi' (measuring the BMI of the patient), and 'diabetes' (recording 0 for patients without diabetes and 1 for patients with diabetes).

Before conducting subsequent feature engineering steps, we first checked for missing values in any of our selected columns. Any rows containing missing values were excluded from the analysis. We expected little to no missing values from our initial screening of the data, and therefore we expected that we had to do little to no handling of missing values. Every row that did not contain a missing value was used for our analysis.

Next, we examined our variables to see if we needed to convert any into categorical variables for our association rule mining step. Out of our three independent variables, 'age' and 'bmi' are continuous variables. First, we split the 'age' variable into 8 separate binary variables. We used the cut function from the pandas library to convert the ages into age ranges: 0 to 10 years old, 11 to 20 years old, 21 to 30 years old, 31 to 40 years old, 41 to 50 years old, 51 to 60 years old, 61 to 70 years old, and 71 to 80 years old. These age ranges cover all of the ages in our data, since we sorted our dataset by 'age' to discover that the youngest age is 0.08 years old and the oldest age is 80 years old. We used the get_dummies function from pandas to generate 8 dummy variables, and for each, a value of 1 was assigned if the patient's age was within the

labeled age range, and a value of 0 was assigned otherwise. Second, we split the 'bmi' variable into 6 separate binary variables. Each of these new variables represented a BMI range in our data (measured in kg/m^2) and were labeled by these BMI ranges with the cut function: 10 to 25, 26 to 40, 41 to 55, 56 to 70, 71 to 85, and 86 to 100. We had sorted our dataset by 'bmi' to find that the lowest BMI value was 10.01, and that the highest was 95.69. We used the `get_dummies` function to generate the 6 dummy variables, and for each, a value of 1 was assigned if the patient's BMI was within the labeled range, and a value of 0 was assigned otherwise. This concluded preparing our continuous variables for the association rule mining step.

Afterwards, we created dummy variables for our 'gender' variable. The purpose of this was twofold: these new variables would be in an appropriate format to be used both for the association rule mining step, which requires the variables to be split into 1's and 0's, and for the logistic regression step, which benefits from having categorical variables converted into a numerical format. Prior to making our dummy variables, we eliminated the rows where the gender was listed as "Other" due to being too ambiguous for our analysis and having analytical value equivalent to a missing value. Afterwards, two separate dummy variables were generated from our 'gender' variable with the `get_dummies` function. One dummy variable, labeled 'gender_Male', was assigned a value of 1 if the patient was male and a value of 0 if the patient was female, while the second dummy variable, labeled 'gender_Female', was assigned a value of 1 if the patient was female and a value of 0 if the patient was male. The first dummy variable was designed to test for an association between males and the risk of diabetes, while the second was designed to test for an association between females and the risk of diabetes. Because it would not matter which dummy variable we chose for our logistic regression modeling step, we simply chose to use the first dummy variable for our modeling step.

Association Rule Mining

After preparing the data for analysis, we began our analysis by examining the relationships between our potential risk factors and the presence of diabetes. The purpose of this step was to see if there was any association between our potential risk factors and the presence of diabetes, and also to examine any association between our potential risk factors themselves, in the case that grouping certain risk factors together may increase or decrease the risk of diabetes.

We achieved this purpose by finding rules that predict a particular item or event's occurrence based on the other occurrences that occur simultaneously. In our case, the antecedents in our rules were the risk factors of interest, while the consequent was the presence of diabetes. We imported the mlxtend library in Python, and from the frequent_patterns module in the library, we used the apriori and association_rules functions. The minimum thresholds for support and confidence rate were both set to 0.01. For a particular rule, in our case, the support measures the fraction of the total data points that contain both the risk factor(s) of interest and the positive presence of diabetes. The confidence measures the fraction of data points containing the potential risk factor(s) that also contain the positive presence of diabetes. Higher support and confidence values will indicate a stronger association between the risk factor(s) and the presence of diabetes, which is of interest to us for our objective of measuring the relationships between the risk factors and diabetes. For the final phase of our association rule mining step, we analyzed the generated rules with the highest support and confidence values to draw tentative conclusions about which risk factors or groupings of risk factors were most associated with the presence of diabetes.

Logistic Regression

In addition to the analysis done by the association rule mining, we used logistic regression to assess how strong the association between each risk factor and the presence of diabetes was. This required importing the sklearn module and using the linear_model class, which contains the function LogisticRegression. Logistic regression is an analytical technique that can be used to examine the relationship between an independent variable and a binary dependent variable, making it ideal in our case, where we wish to examine the relationship between the risk factors and our binary variable of whether or not a patient has diabetes.

To begin the regression analysis, we split our data into training and test sets. This required importing the train_test_split function from the sklearn.model_selection module. Using the train_test_split function, we split every variable, consisting of the 3 potential risk factors and the presence of diabetes, into a training set that consisted of 80% of the data and a test set that consisted of 20% of the data. The purpose of randomly splitting the data in this fashion was to use the training data for our predictive models, and then to test the model against our test data to

evaluate how accurate the model was at using the data of the risk factors to predict whether someone has diabetes or not.

After splitting the data into training and test sets, we generated four predictive models. Three of the models were created using the LogisticRegression function and fitting the models to three different pairs: each pair consisted of one risk factor and the presence of diabetes. The purpose of these three initial models was to assess how accurately diabetes could be predicted using a single risk factor as a predictor. Having a high accuracy with just a single predictor would indicate a strong association between the risk factor and the presence of diabetes, which would lead to suggestions for a greater focus on the risk factor when planning preventive measures and intervention strategies for diabetes. We generated predictions of diabetes based on each model with the predict function in Python, with the purpose of later using the accuracy_score function from the sklearn.metrics module to test the predictions against the test data of the presence of diabetes. We would later compare the accuracies of these models with each other and assess the accuracy scores to determine each model's utility.

The fourth model was created using the LogisticRegression function as well, but the model was fitted to all three risk factors in relation to the presence of diabetes. This used a more specific version of logistic regression called multiple logistic regression, which uses several independent variables in relation to one binary dependent variable. The purpose of this final model was to assess how accurately the presence of diabetes could be predicted using all three of our risk factors as predictors. The benefit of this approach would be that there would hopefully be an increased accuracy in our ability to predict diabetes compared to when we use only a single predictor, while the downside of this approach would be that it would be difficult to highlight which risk factors most contributed to being able to predict diabetes. As before, we generated predictions of diabetes based on the model with the predict function, in order to later test these predictions against the test data of the presence of diabetes. We would use these accuracy scores to test how well this more holistic model performed relative to the three initial models.

Testing Predictive Models

In our last step, we explored several metrics that are key to assessing the model's performance, including accuracy score, sensitivity, specificity, false positive rate, and false

negative rate. Using the `accuracy_score` function from the `sklearn.metrics` module, we tested the predictions that we generated against the test data of the presence of diabetes. We generated the accuracy scores for each model as a percentage, by multiplying the original accuracy scores by 100. These scores gave us a general understanding of how accurate each model was, as well as a general understanding of the relative accuracy of each model.

Next, we used the `confusion_matrix` function to produce a confusion matrix for each model. The confusion matrix is a table that helps assess the performance of a model by displaying the count of correct predictions for both outcomes of the binary dependent variable, as well as the count of incorrect predictions for both outcomes of the binary dependent variable. The confusion matrix allowed us to calculate several key metrics related to model performance. First, it allowed us to calculate the sensitivity of our model, which is a measure of the model's capability for accurately predicting true positives, calculated as the percentage of observed positive outcomes that were correctly predicted by the model. Second, it allowed us to calculate the specificity of our model, which is a measure of the model's capability for accurately predicting true negatives, calculated as the percentage of observed negative outcomes that were correctly predicted by the model. Third, the confusion matrix allowed us to calculate the false positive rate, which is a measure of the model's inaccuracy, and calculated as the percentage of observed negative outcomes that were falsely predicted. Fourth, the confusion matrix allowed us to calculate the false negative rate, which is also a measure of the model's inaccuracy, and calculated as the percentage of observed positive outcomes that were falsely predicted. Lastly, the confusion matrix allowed us to calculate the misclassification rate, a final measure of the model's inaccuracy, and calculated as the percentage of all the predictions that were correctly predicted. These metrics helped to have a holistic understanding of each model's overall accuracy and performance, and to make suggestions about each model's use cases.

Finally, we generated ROC (Receiver-Operating Characteristic) curves for the models to display and assess each model's performance visually. The AUC value, or area under the curve, for each ROC curve is a number between 0.5 and 1 that gives an indication about the model's performance. A higher AUC corresponds to a better model performance, with a score of 1 representing a perfectly fitting model and a score of 0.5 representing a model with a performance no better than a random model. The ROC curves were created using the `roc_curve` function from the `sklearn.metrics` module on the test data and the generated predictions to determine sensitivity

and false positive rates at different threshold values. The auc function from the sklearn.metrics module was used to calculate each ROC curve's AUC value based on the sensitivity and false positive rate values generated by the roc_curve function. Afterwards, matplotlib.pyplot was used to plot the ROC curves, with the false positive rate as the x-axis and the sensitivity value as the y-axis. The AUC value of each ROC curve accompanied each ROC curve, giving insight into how well the model performed when fitted to the training data. The AUC values were used to assess the model's relative performance, as well as its performance based on general standards on what an AUC value indicates about a model's performance.

Results

Data Understanding

Our exploration of the Diabetes Prediction dataset allowed us to have a rudimentary understanding of the relationships between the risk factors and the presence of diabetes prior to our analysis. We also gained insight into the characteristics of each variable through their key statistics and through visualization of individual variables and the relationships that the independent variables have with the target variable.

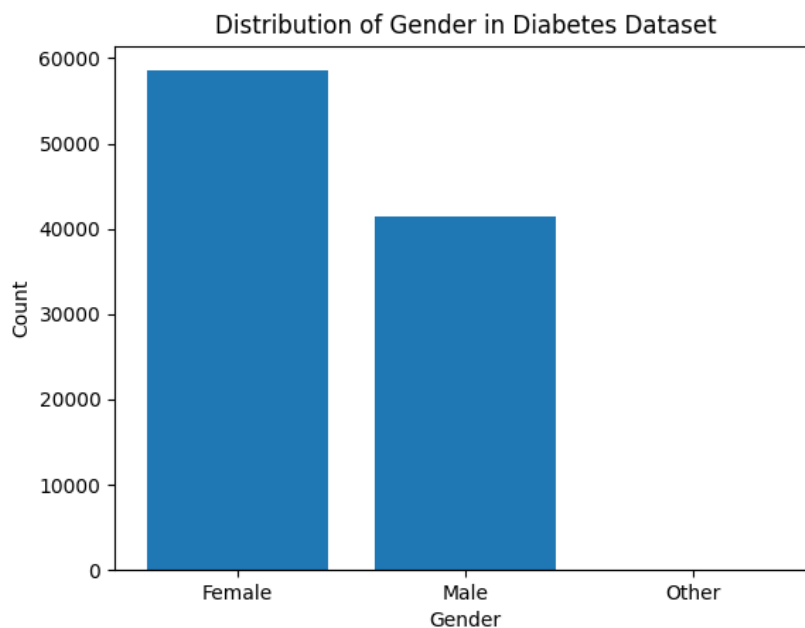
Figure 1 – Key Statistics for Age, BMI, and Presence of Diabetes

	age	bmi	diabetes
count	100000.000000	100000.000000	100000.000000
mean	41.885856	27.320767	0.085000
std	22.516840	6.636783	0.278883
min	0.080000	10.010000	0.000000
25%	24.000000	23.630000	0.000000
50%	43.000000	27.320000	0.000000
75%	60.000000	29.580000	0.000000
max	80.000000	95.690000	1.000000

In this figure, we see the key statistics for the age, BMI, and diabetes variables. Notably, the gender variable was excluded since it is a non-numerical variable. We see that the average age is around 42 years of age, with a standard deviation of around 22.5 years of age, which is relatively large. The ages range from 0.08 years to 80 years, justifying our need to categorize the data in

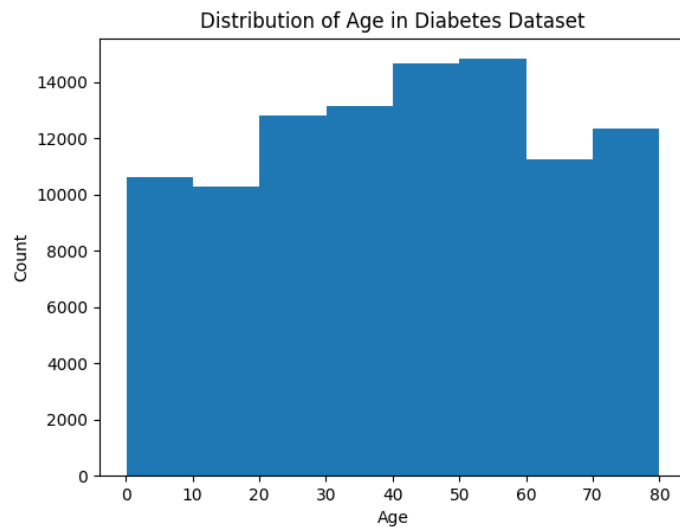
our data preparation step into age ranges starting from 0 years of age. Next, we see that the average BMI is around 27.32, with a standard deviation of roughly 6.64. The BMI values range from 10.01 to 95.69, justifying our decision to categorize the data in our data preparation step into BMI ranges starting from a value of 10. Lastly, although it is a binary variable split into 0's and 1's, we note that the mean value for the diabetes variable is 0.085, indicating that the majority of the data points possess a value of 0 to represent an absence of diabetes. These statistics give insight into the characteristics of these three variables, and allow us to plan for our data preparation step.

Figure 2 – Histogram of Gender



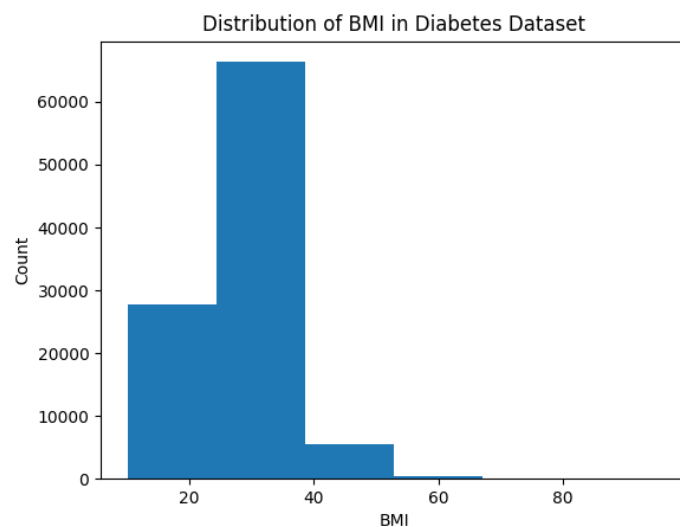
In this figure, we see a histogram displaying the distribution of the genders in our dataset. There are close to 20,000 more females recorded than males, with a negligible number of individuals being recorded as “Other”. We conclude from this figure that there are enough of both males and females to analyze the differences between the two genders in predicting diabetes.

Figure 3 – Histogram of Age



In this figure, we see a histogram displaying the distribution of the ages recorded in our dataset. Each bin represents an age range of 10 years starting from 0 years of age. We note that none of our bins is abnormally higher or lower than the others, leading us to conclude that it is appropriate to divide this variable into these 8 age ranges for our analysis.

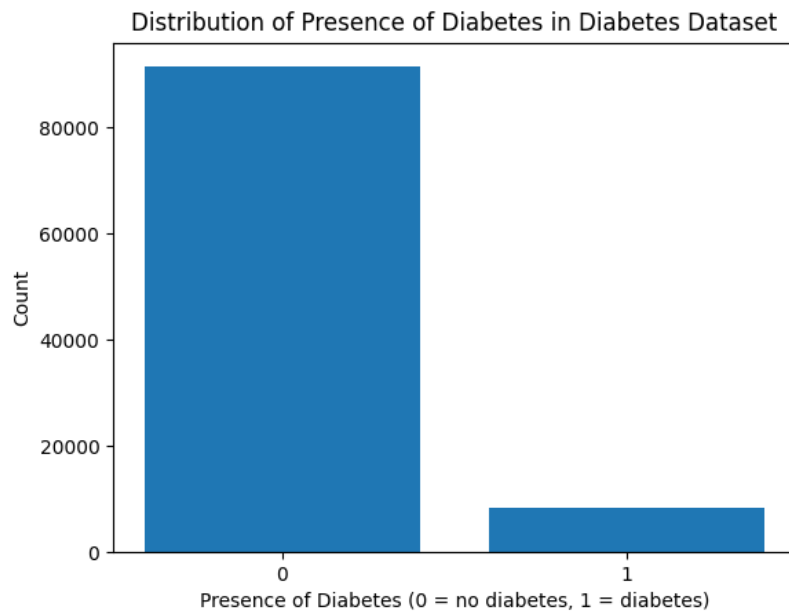
Figure 4 – Histogram of BMI



In this figure, we see a histogram displaying the distribution of the BMI values recorded in our dataset, split into 6 bins. We note that there is a strong positive skewness in this distribution, with most of the values concentrated from roughly 25 to 40, and there being relatively few higher values. We will have to take the skewness into consideration in our analysis, since it may affect

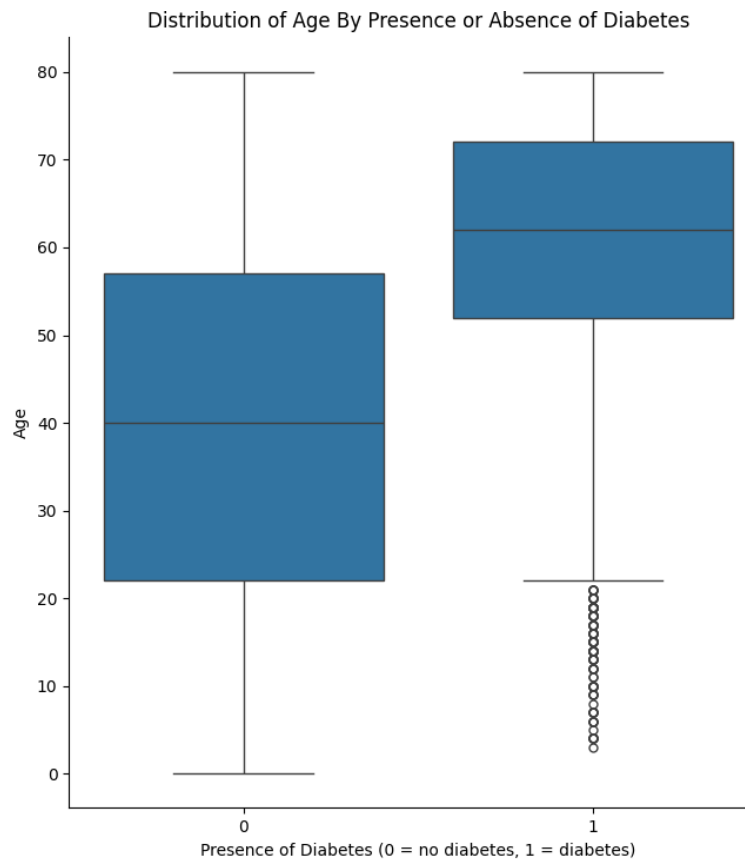
how accurate the association rule mining may be with a small sample size in the higher-end bins. The skewness should not affect logistic regression since logistic regression does not assume normality.

Figure 5 – Histogram of Presence of Diabetes



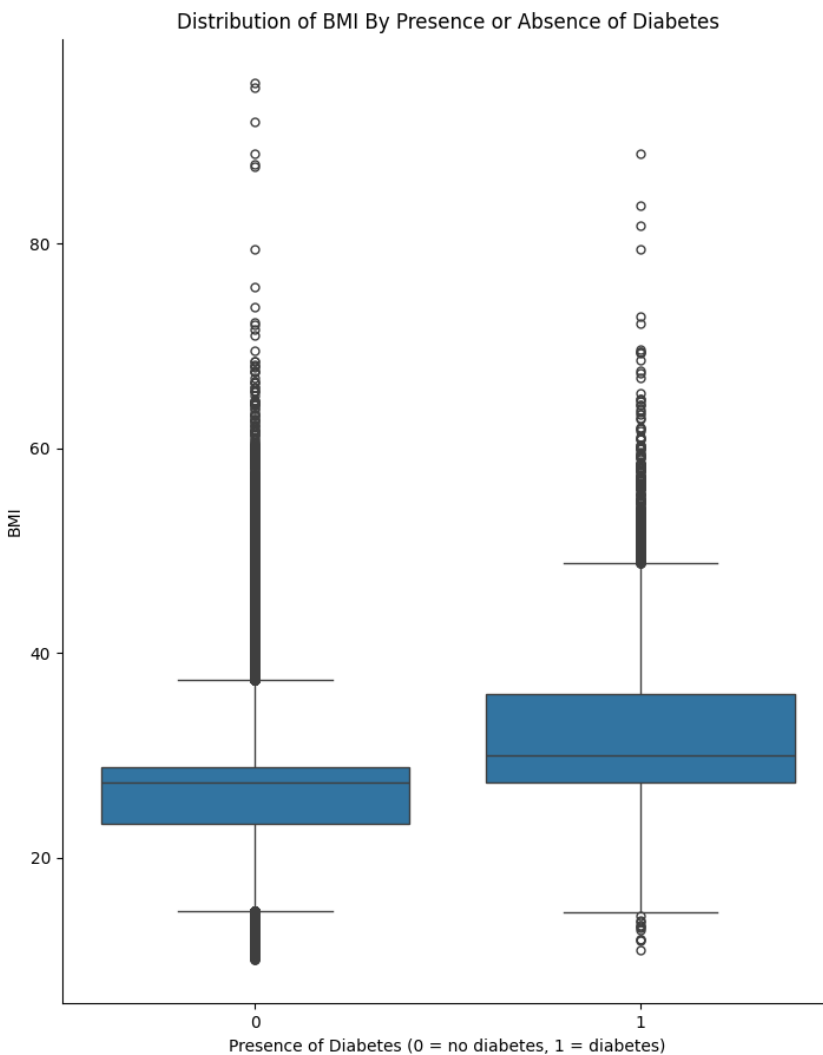
In this figure, we see a histogram displaying the distribution of our binary target variable, where a value of 0 indicates that the individual does not have diabetes, while a value of 1 indicates that the individual has diabetes. We note from this figure that there are significantly more data points where there is an absence of diabetes than data points where there is diabetes. However, we also observe that there are thousands of data points where the individual does have diabetes, and we determine that this is a sufficient sample size for analyzing the relationship between diabetes and its risk factors.

Figure 6 – Boxplot of Diabetes vs. Age



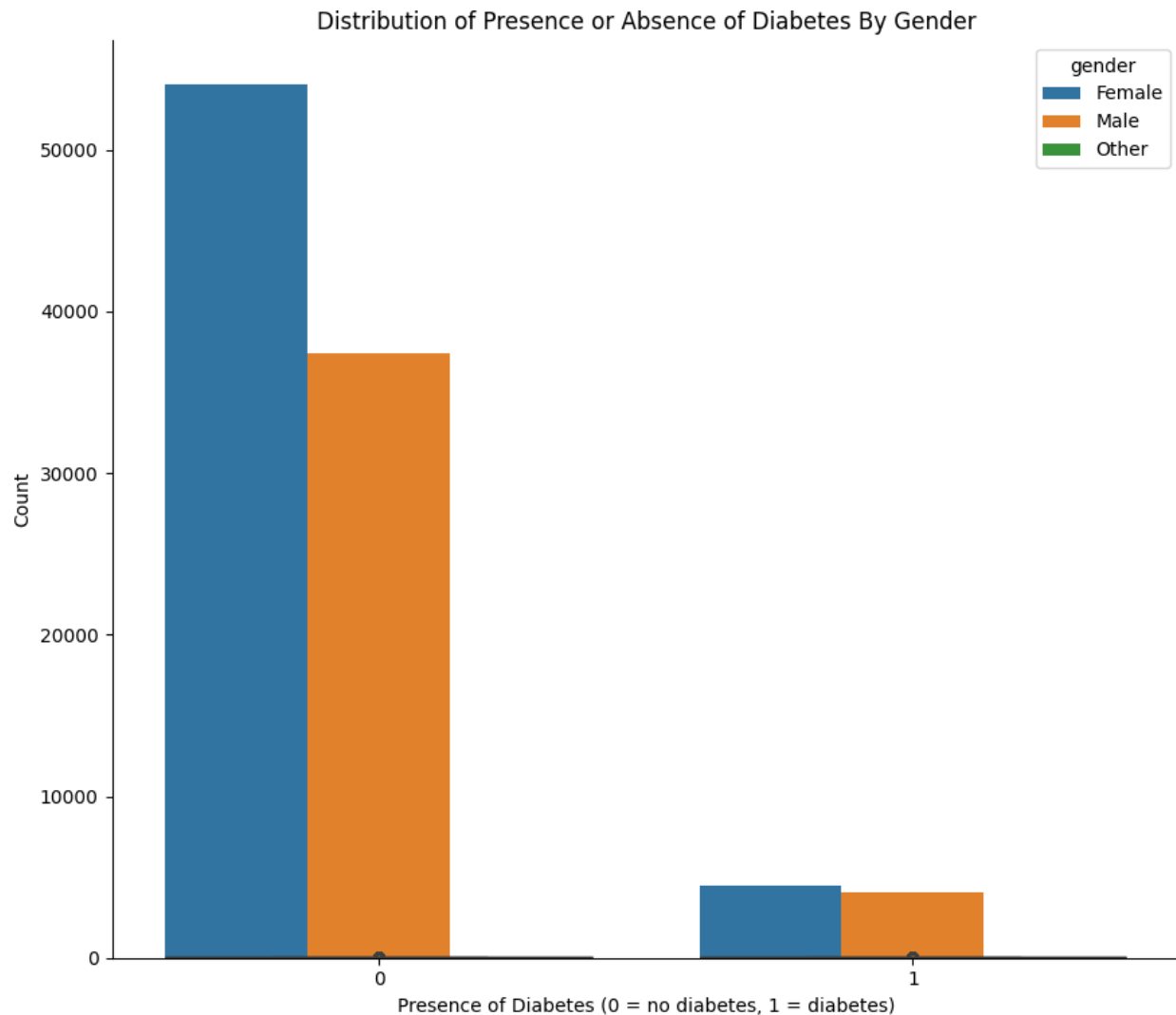
In this figure, we see a boxplot of the distribution of the ages in our dataset for those with diabetes and for those without diabetes. Through visual examination, it seems clear that in general, the age of someone with diabetes tends to be higher than the age of someone without diabetes, indicating that the risk of diabetes may increase with age. The median age for those without diabetes is roughly 40 years old, while the median age for those with diabetes is a little over 60 years old. Although we will draw more robust conclusions after the totality of our analysis, this initial insight into the relationship between age and the presence of diabetes seems to call for a heightened emphasis on preventive care for those who are older, since they are seemingly more at risk of developing diabetes.

Figure 7 – Boxplot of Diabetes vs. BMI



In this figure, we see a boxplot of the distribution of the BMI values in our dataset for those with diabetes and for those without diabetes. Through visual examination, we can tentatively note that the BMI of someone with diabetes tends to be higher than the BMI of someone without diabetes, indicating that the risk of diabetes may increase with BMI. The median BMI value for those without diabetes is notably less than the median BMI value for those with diabetes, and the upper quartile of the BMI values of those without diabetes is also notably less than the upper quartile of the BMI values of those with diabetes. This initial insight into the relationship between BMI and the presence of diabetes seems to call for a preventive measure where individuals are encouraged to adopt lifestyles that would reduce their BMI values, and thus correspondingly decrease their potential risk of developing diabetes. Individuals with high BMI values may have to be placed under special care for reducing their BMI values in an attempt to avoid diabetes development.

Figure 8 – Bar Chart of the Presence of Diabetes by Gender



In this figure, we see a bar chart displaying the distribution of the diabetes data sorted by gender. Similarly to Figure 2, we note that the gender labeled “Other” is relatively negligible in the data distribution. We observe that there are roughly 15,000 more females than males out of those who do not have diabetes, but that the number of females and males seem roughly equal out of those who do have diabetes. This may indicate that gender is not a significant predictor of diabetes development, which could alter existing strategies that have prioritized one gender over the other in the deployment of preventive measures for diabetes. We will draw more credible conclusions after completing our analysis, but this initial insight is helpful for having at least an elementary understanding of the relationship between gender and the presence of diabetes.

Data Preparation and Feature Engineering

We proceeded to prepare our data for the association rule mining and logistic regression steps. As mentioned previously, we converted our continuous independent variables, 'age' and 'bmi', into categorical variables that split 'age' and 'bmi' into ranges. We then generated dummy variables from our new categorical variables, where the labeled range was assigned a value of 1, and all other ranges were assigned a value of 0. The 'age' variable was converted into a categorical variable called 'age_range' containing the age ranges of 0 to 10 years old, 11 to 20 years old, 21 to 30 years old, 31 to 40 years old, 41 to 50 years old, 51 to 60 years old, 61 to 70 years old, and 71 to 80 years old, and was then split into 8 dummy variables. The 'bmi' variable was converted into a categorical variable called 'bmi_range' containing the BMI ranges of 10 to 25, 26 to 40, 41 to 55, 56 to 70, 71 to 85, and 86 to 100, and was then split into 6 dummy variables. These dummy variables would be used in the association rule mining step to associate each age or BMI range with the presence of diabetes.

We also generated two dummy variables for our 'gender' variable, with one of them having a value of 1 for males and a value of 0 for females, and the other dummy variable having a value of 1 for females and a value of 0 for males. Prior to generating these two dummy variables, we eliminated all the rows in our data where the gender was labeled as 'Other' due to this value being too ambiguous and having the analytical value of a missing value. These dummy variables would be useful in the association rule mining step, where each gender could be associated with the presence of diabetes. The dummy variable for males would be used in the logistic regression step to associate the 'gender' variable with the presence of diabetes.

The resulting dataset had 99,982 rows and 19 columns after generating the dummy variables and eliminating the rows with a gender value of 'Other'.

Figure 9 – Data Types of Variables After Feature Engineering

age	float64
bmi	float64
diabetes	int64
age_range_0-10	int64
age_range_11-20	int64
age_range_21-30	int64
age_range_31-40	int64
age_range_41-50	int64
age_range_51-60	int64
age_range_61-70	int64
age_range_71-80	int64
bmi_range_10-25	int64
bmi_range_26-40	int64
bmi_range_41-55	int64
bmi_range_56-70	int64
bmi_range_71-85	int64
bmi_range_86-100	int64
gender_Female	int64
gender_Male	int64

This figure shows our new list of 19 variables after the dummy variables were generated, and after the ‘gender’ variable was split into two dummy variables for male and female. Every variable below ‘age’ and ‘bmi’ are of the int-type since they are categorized by 0’s and 1’s.

Association Rule Mining

Our association rule mining step was performed with the presence of diabetes being the sole consequent. For the sake of convenience, a new dataset was generated that removed the ‘age’ and ‘bmi’ variables, leaving only the ‘diabetes’ variable and the 16 dummy variables that were generated in the previous step. This left only the binary variables that were categorized by 0’s and 1’s for our association rule mining step.

Figure 10 – Association Rule Mining Table

	antecedents	consequents	support	confidence
0	(bmi_range_41-55)	(diabetes)	0.010832	0.252801
1	(gender_Male, age_range_71-80)	(diabetes)	0.011452	0.226151
2	(gender_Male, age_range_61-70)	(diabetes)	0.010852	0.224964
3	(age_range_71-80, bmi_range_26-40)	(diabetes)	0.019484	0.212224
4	(age_range_71-80)	(diabetes)	0.024904	0.201522
5	(bmi_range_26-40, age_range_61-70)	(diabetes)	0.017543	0.198641
6	(age_range_71-80, bmi_range_26-40, gender_Female)	(diabetes)	0.010282	0.198532
7	(age_range_61-70)	(diabetes)	0.022294	0.197676
8	(age_range_71-80, gender_Female)	(diabetes)	0.013452	0.184423
9	(gender_Female, age_range_61-70)	(diabetes)	0.011442	0.177282
10	(age_range_51-60)	(diabetes)	0.019313	0.130227
11	(bmi_range_26-40, age_range_51-60)	(diabetes)	0.014403	0.127074
12	(gender_Male, bmi_range_26-40)	(diabetes)	0.031926	0.112122
13	(bmi_range_26-40)	(diabetes)	0.063821	0.098805
14	(gender_Male)	(diabetes)	0.040397	0.097490
15	(bmi_range_26-40, gender_Female)	(diabetes)	0.031896	0.088306
16	(gender_Female)	(diabetes)	0.044618	0.076189
17	(age_range_41-50)	(diabetes)	0.010692	0.072910

In this figure, we see a table generated after applying association rule mining to our dataset, with a minimum support and confidence threshold of 0.01. The antecedents, support values, and confidence values are displayed, with the consequent being set to the presence of diabetes. We observed that the support values are quite low, which is to be expected, since we knew from Figure 5 that there are far less data points with a positive outcome for diabetes than data points with a negative outcome for diabetes. Therefore, because the fraction of data points with a positive value for diabetes is already low, the fraction of data points that contain both the antecedent(s) and a positive value for diabetes must also be low.

On the other hand, the confidence values are much higher, and give insight into which risk factors may be more strongly associated with the presence of diabetes. The antecedent with the highest confidence value is a BMI range of 41 to 55: roughly 25.28% of the patients with a BMI between 41 and 55 also had diabetes. This indicates that individuals with a higher BMI may be at a higher risk of developing diabetes. We notice from the second and the third highest confidence values (22.62 and 22.50) that males with ages in the range 61-70 are relatively more associated with the presence of diabetes, indicating that elderly males may be at a higher risk of diabetes.

Additionally, every data point from the second to the twelfth in terms of highest confidence value consists of an age range that is between 51 and 80, giving a strong indication that the elderly are more susceptible to developing diabetes. Based on Figure 4, we can likely afford to discount the occurrence of the 26-40 BMI range in our association rule mining table, since it makes up the majority of our data points and likely occurs due to being common rather than from having a strong association with diabetes. Overall, it seems that diabetes is most associated with a higher BMI range of 41 to 55, a higher age range of roughly 51 to 80, and potentially with males more than with females.

Logistic Regression

For our logistic regression analysis, we generated four models for predicting the presence of diabetes:

1. The Gender Model, which used the 'gender' variable, or more specifically the 'gender_Male' variable, to predict the presence of diabetes in an individual.
2. The Age Model, which used the 'age' variable to predict the presence of diabetes in an individual.
3. The BMI Model, which used the 'bmi' variable to predict the presence of diabetes in an individual.
4. The Full Model, which used the 'gender', 'age', and 'bmi' variables to predict the presence of diabetes in an individual.

Each variable was split into a training set that consisted of 80% of the data in the variable, and a test set that consisted of 20% of the data in the variable. A logistic regression model was fit to the appropriate training sets (the predictor(s) and the diabetes data), and then tested against the test set for each model. We will take a look at the accuracy of each model through several metrics in the final step.

Testing Predictive Models

For our final step of our analysis, we analyzed the models generated through our logistic regression method. We extracted and calculated several key statistics that helped us to assess the accuracy and performance of each model.

Figure 11 – Model Accuracy Scores and Confusion Matrices

```
Gender Model Accuracy is 91.62374356153423%
[[ 0 1675]
 [ 0 18322]]
Age Model Accuracy is 91.44871730759614%
[[ 0 1710]
 [ 0 18287]]
BMI Model Accuracy is 91.04365654848228%
[[ 26 1745]
 [ 46 18180]]
Full Model Accuracy is 91.74876231434716%
[[ 50 1598]
 [ 52 18297]]
```

In this figure, we generated the accuracy scores of each model using the in-built `accuracy_score` function from the `sklearn.metrics` module, and generated the confusion matrix for each model.

We note that the accuracy was roughly 91.62% for the Gender Model, 91.45% for the Age Model, 91.04% for the BMI Model, and 91.75% for the Full Model. We observed that the accuracy scores are all high, with prediction accuracies above 90% for each of them, and that the accuracy scores are all similar to one another. Although the difference may be negligible, we note that that accuracy of the Full Model is the highest, which is what we anticipated, since it draws from information provided by all three risk factors rather than from a single one of them.

We then analyzed the confusion matrices associated with the models. Each confusion matrix had the number of true positives in the top-left corner, the number of false negatives in the top-right corner, the number of false positives in the bottom-left corner, and the number of true negatives in the bottom-right corner. Using the confusion matrices, we calculated the sensitivity, specificity, false positive rate, false negative rate, and misclassification rate for each model. For

our calculations, TP = number of true positives, TN = number of true negatives, FP = number of false positives, and FN = number of false negatives.

1. Gender Model

- a. Sensitivity = $TP / (TP + FN) = 0 / (0 + 1675) = 0$ or 0%
- b. Specificity = $TN / (FP + TN) = 18322 / (0 + 18322) = 1$ or 100%
- c. False Positive Rate = $FP / (TN + FP) = 0 / (18322 + 0) = 0$ or 0%
- d. False Negative Rate = $FN / (TP + FN) = 1675 / (0 + 1675) = 1$ or 100%
- e. Misclassification Rate = $(FP + FN) / (TP + TN + FP + FN) = (0 + 1675) / (0 + 18322 + 0 + 1675) = 0.0838$ or 8.38%

2. Age Model

- a. Sensitivity = $TP / (TP + FN) = 0 / (0 + 1710) = 0$ or 0%
- b. Specificity = $TN / (FP + TN) = 18287 / (0 + 18287) = 1$ or 100%
- c. False Positive Rate = $FP / (TN + FP) = 0 / (18287 + 0) = 0$ or 0%
- d. False Negative Rate = $FN / (TP + FN) = 1710 / (0 + 1710) = 1$ or 100%
- e. Misclassification Rate = $(FP + FN) / (TP + TN + FP + FN) = (0 + 1710) / (0 + 18287 + 0 + 1710) = 0.0855$ or 8.55%

3. BMI Model

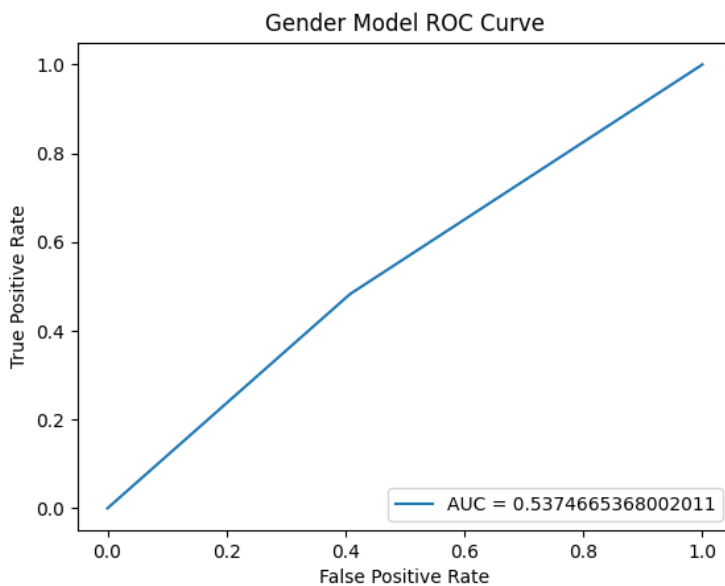
- a. Sensitivity = $TP / (TP + FN) = 26 / (26 + 1745) = 0.0147$ or 1.47%
- b. Specificity = $TN / (FP + TN) = 18180 / (46 + 18180) = 0.9975$ or 99.75%
- c. False Positive Rate = $FP / (TN + FP) = 46 / (18180 + 46) = 0.0025$ or 0.25%
- d. False Negative Rate = $FN / (TP + FN) = 1745 / (26 + 1745) = 0.9853$ or 98.53%
- e. Misclassification Rate = $(FP + FN) / (TP + TN + FP + FN) = (46 + 1745) / (26 + 18180 + 46 + 1745) = 0.0896$ or 8.96%

4. Full Model

- a. Sensitivity = $TP / (TP + FN) = 50 / (50 + 1598) = 0.0303$ or 3.03%
- b. Specificity = $TN / (FP + TN) = 18297 / (52 + 18297) = 0.9972$ or 99.72%
- c. False Positive Rate = $FP / (TN + FP) = 52 / (18297 + 52) = 0.0028$ or 0.28%
- d. False Negative Rate = $FN / (TP + FN) = 1598 / (50 + 1598) = 0.9697$ or 96.97%
- e. Misclassification Rate = $(FP + FN) / (TP + TN + FP + FN) = (52 + 1598) / (50 + 18297 + 52 + 1598) = 0.0825$ or 8.25%

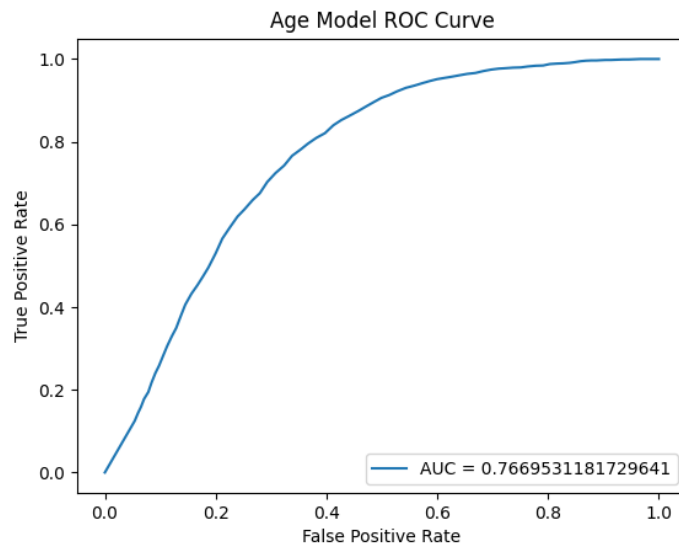
We note that for all of our models, the specificity and false negative rates are all extremely high, between 96.97% and 100%, while the sensitivity and false positive rates are extremely low, between 0% and 3.03%. The misclassification rates are all around 8%, with the lowest being 8.25%, for the full model. Because the misclassification rate is the lowest for the full model, and since the full model also had the highest accuracy score, we would select the full model as our chosen model for predicting diabetes as of now.

Figure 12 – Gender Model ROC Curve



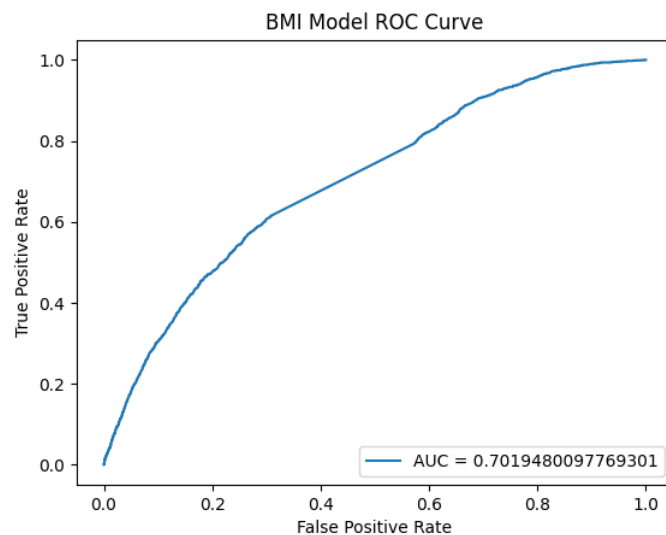
In this figure, we see the ROC curve for the Gender Model. We observed that the AUC is roughly 0.537, which is close to the lowest possible AUC value of 0.5. This indicates that the model had nearly no discriminatory power, meaning that the model performed barely better than random guesswork.

Figure 13 – Age Model ROC Curve



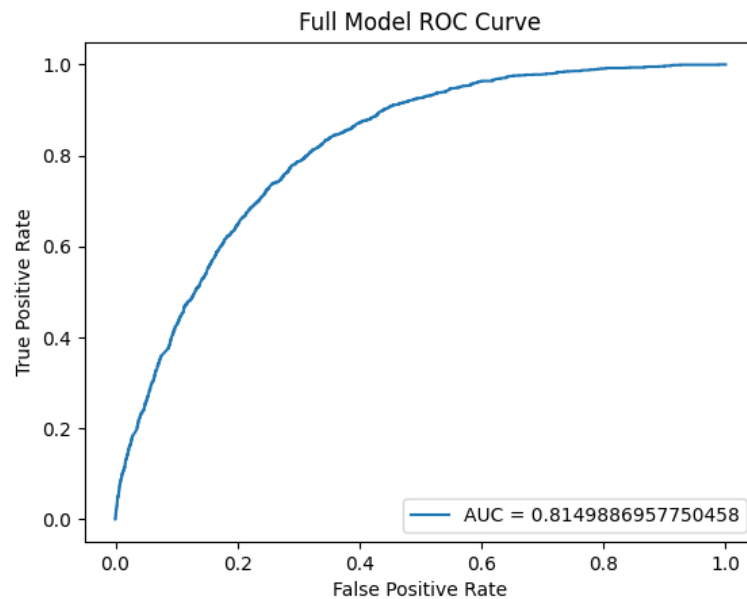
In this figure, we see the ROC curve for the Age Model. We observed that the AUC is roughly 0.767, which is roughly halfway between the lowest possible AUC value of 0.5 and the highest possible AUC value of 1. This indicates that the model had a moderate amount of discriminatory power, meaning that the model had an acceptable classification performance.

Figure 14 – BMI Model ROC Curve



In this figure, we see the ROC curve for the Gender Model. We observed that the AUC is roughly 0.702, which is near the middle of the lowest possible AUC value of 0.5 and the highest possible AUC value of 1. This indicates that the model had a moderate amount of discriminatory power, meaning that the model had an acceptable classification performance.

Figure 15 – Full Model ROC Curve



In this figure, we see the ROC curve for the Gender Model. We observed that the AUC is roughly 0.815, which is relatively close to the highest possible AUC value of 1. This indicates that the model had a relatively high amount of discriminatory power, meaning that the model had an excellent classification performance, and the best performance out of our four models. This further justifies our decision to select the full model as our chosen model to predict diabetes.

Conclusion

Discussion of Results

We drew several conclusions from our analytical results. From Figure 10, where we displayed the results of our association rule mining analysis, we observed that the scenario that was most associated with the presence of diabetes was when an individual had a BMI of 41 to 55, which would be a high BMI value. Based off of this information, we would agree with all of our literature on diabetes that increased BMI puts an individual at an increased risk of diabetes. We would suggest that preventive measures be strengthened to reduce the BMI of those who are significantly above a standard BMI value.

We also observed from Figure 10 that higher age ranges were more strongly associated with the presence of diabetes. The age ranges of 71 to 80 years old and 61 to 70 years old were

strongly associated with diabetes, with the age range of 51 to 60 years old being the next strongly associated with diabetes. Thus, we conclude that older individuals are more at risk of diabetes, in line with many of the research studies that have been done on the relationship between age and diabetes risk (Chen , 2018). This conclusion contradicts the findings of some research studies that have observed that those who are pre-diabetic and thus at a greater risk of diabetes are often younger than those who do are not pre-diabetic (Chaudhuri, 2024). We would suggest that further study be done on the strength of age as a predictor of diabetes, and place an increased emphasis on providing preventive care to those who are in the age range that is more at risk of diabetes.

We noticed from Figure 10 that older males may be more strongly associated with diabetes than any females. This would suggest that males may be more at risk of diabetes, which would contradict the conclusions of studies that have deemed that females are at an increased risk of diabetes (Kapur, 2017). However, this analytical result is not definitive, and does not address the reality that females are likely to face greater complications with diabetes, especially during pregnancy, and therefore one gender should not be preferred to the other in the administration of preventive care.

The statistics generated for our logistic regression models, which included sensitivity, specificity, false positive rate, and false negative rate indicated that our models all possessed high specificity and false negative rates, but very low sensitivity and false positive rates. This implied that the models perform well in detecting true negative outcomes and avoiding false positive outcomes, but perform poorly in detecting true positive outcomes and avoiding false negative outcomes. We suspect that these statistics may be skewed due to having too low of a sample size for individuals with diabetes in comparison with the sample size for individuals without diabetes.

The models were tested more accurately with the ROC curves and AUC values in Figures 12 to 15. We observed that the Gender Model had a low AUC value, the Age Model and BMI Model had moderate AUC values, and the Full Model had a high AUC value. This suggests that gender is not likely to be a significant predictor of diabetes, and may not be strongly associated with diabetes. Age and BMI are both risk factors that would be moderately strong predictors of diabetes, which is knowledge that is already widely accepted in the field of diabetes research (Wu, 2014). Using multiple variables to predict the presence of diabetes seems to increase the

predictive accuracy, as shown by the high AUC value for the Full Model, while having potential downsides such as increasing the model's complexity (Smith, 1988).

In summary, taking into account the entirety of our analysis, we conclude that gender is unlikely to be significantly associated with the risk of diabetes, while age and BMI are significant risk factors for diabetes. Individuals who are elderly and individuals with high BMI are likely to be more at risk of diabetes, and therefore preventive measures, such as a change in diet or exercise routine, should be incorporated to protect elderly individuals and to help reduce BMI for individuals with high BMI.

Future Work

Our analysis in this project could lead to subsequent work being done in various ways. Additional models could be generated with increased predictive accuracy by taking into account a longer list of potential risk factors. Different analytical methods, such as decision tree, neural network, and random forest, could be employed to develop models, and these models could be tested against each other to identify a method that is best suited for predicting diabetes. These models could be used to uncover new risk factors, and to further validate the significance of existing risk factors.

Data regarding preventive measures for diabetes and their levels of success at lowering diabetes risk could also be collected and tested in a similar fashion to the way that the risk factors were in this project. Predictive modeling through logistic regression or a different method could be used to identify which preventive measures are most effective at lowering diabetes risk and which ones are ineffective at lowering diabetes risk, so that only the successful preventive measures could be utilized in preventive care for diabetes.

Project Breakdown

Methodologies Learned from MSIS 5193

- Data Understanding
 - Exploring datasets
 - Plotting histograms, box plots, and bar charts

- Data Preparation and Feature Engineering
 - Handling missing values
 - Creating new columns
 - Sorting data
 - Removing rows
- Association Rule Mining

Methodologies Newly Learned

- Logistic Regression
 - Using new Python modules
 - Splitting data into training and test sets
- Testing Predictive Models
 - Using new Python modules
 - Generating confusion matrices
 - Plotting ROC curves and analyzing AUC values

References

Chaudhuri I., Alam M.S., Kamrujjaman M. “Novel Risk Factors for Diabetes: A Comprehensive Analysis for Enhanced Disease Diagnosis.” International Journal of Ground Sediment & Water, 19, 1355–1382 (2024). <https://doi.org/10.5281/zenodo.10791178>.

Chen Y., Zhang X.P., Yuan J., Cai B., Wang X.L., Wu X.L., Zhang Y.H., Zhang X.Y., Yin T., Zhu X.H., Gu Y.J., Cui S.W., Lu Z.Q., Li X.Y. “Association of body mass index and age with incident diabetes in Chinese adults: a population-based cohort study.” BMJ Open 8(9):e021768 (2018). doi:10.1136/bmjopen-2018-021768.

Kapur A., Seshiah V. “Women & diabetes: Our right to a healthy future.” Indian J Med Res 146(5):553-556 (2017). doi: 10.4103/ijmr.IJMR_1695_17.

Smith J.W., Everhart J.E., Dickson W.C., Knowler W.C., Johannes R.S. “Using the ADAP Learning Algorithm to Forecast the Onset of Diabetes Mellitus.” Proc Annu Symp Comput Appl Med Care 261–5 (1988).

Strack B., DeShazo J.P., Gennings C., Olmo J.L., Ventura S., Cios K.J., Clore J.N. “Impact of HbA1c measurement on hospital readmission rates: analysis of 70,000 clinical database patient records.” Biomed Res Int. (2014). doi: 10.1155/2014/781670.

Tigga N.P., Garg S. “Prediction of Type 2 Diabetes using Machine Learning Classification Methods.” Procedia Computer Science, Volume 167, Pages 706-716, ISSN 1877-0509 (2020). <https://doi.org/10.1016/j.procs.2020.03.336>.

Wu Y., Ding Y., Tanaka Y., Zhang W. “Risk factors contributing to type 2 diabetes and recent advances in the treatment and prevention.” Int J Med Sci.;11(11):1185-200 (2014). doi:10.7150/ijms.10001.

Xie Z., Nikolayeva O., Luo J., Li D. “Building Risk Prediction Models for Type 2 Diabetes Using Machine Learning Techniques.” Prev Chronic Dis 16:190109 (2019). <http://dx.doi.org/10.5888/pcd16.190109>

Appendix

Figure 1 – Key Statistics for Age, BMI, and Presence of Diabetes

	age	bmi	diabetes
count	100000.000000	100000.000000	100000.000000
mean	41.885856	27.320767	0.085000
std	22.516840	6.636783	0.278883
min	0.000000	10.010000	0.000000
25%	24.000000	23.630000	0.000000
50%	43.000000	27.320000	0.000000
75%	60.000000	29.580000	0.000000
max	80.000000	95.690000	1.000000

Figure 2 – Histogram of Gender

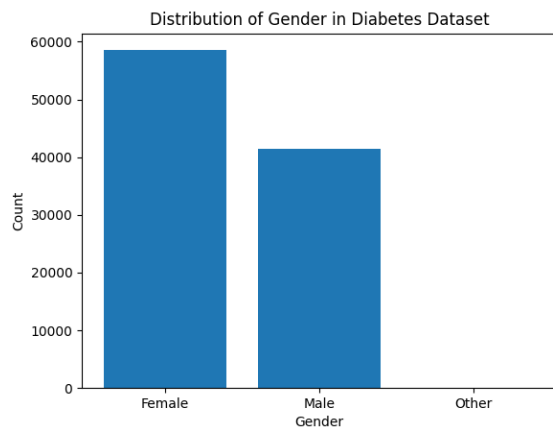


Figure 3 – Histogram of Age

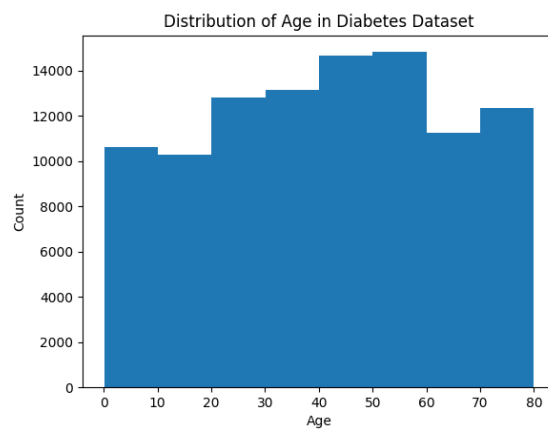


Figure 4 – Histogram of BMI

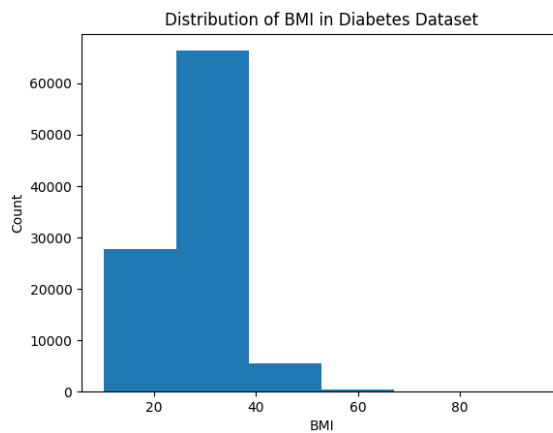


Figure 5 – Histogram of Presence of Diabetes

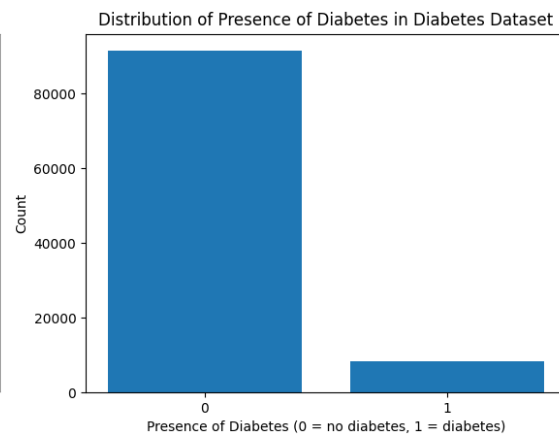


Figure 6 – Boxplot of Diabetes vs. Age

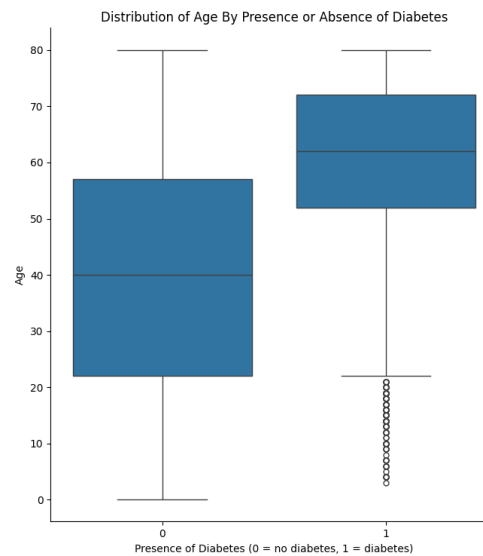


Figure 7 – Boxplot of Diabetes vs. BMI

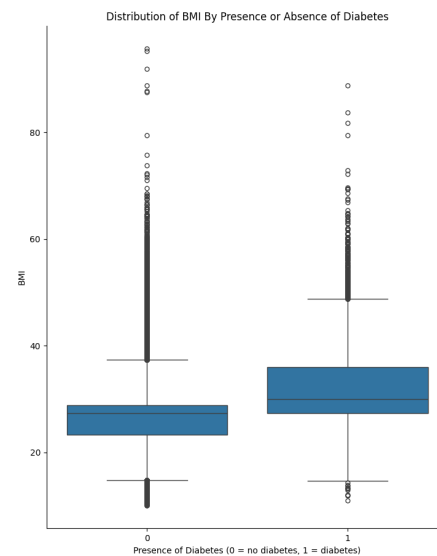


Figure 8 – Bar Chart of the Presence of Diabetes by Gender

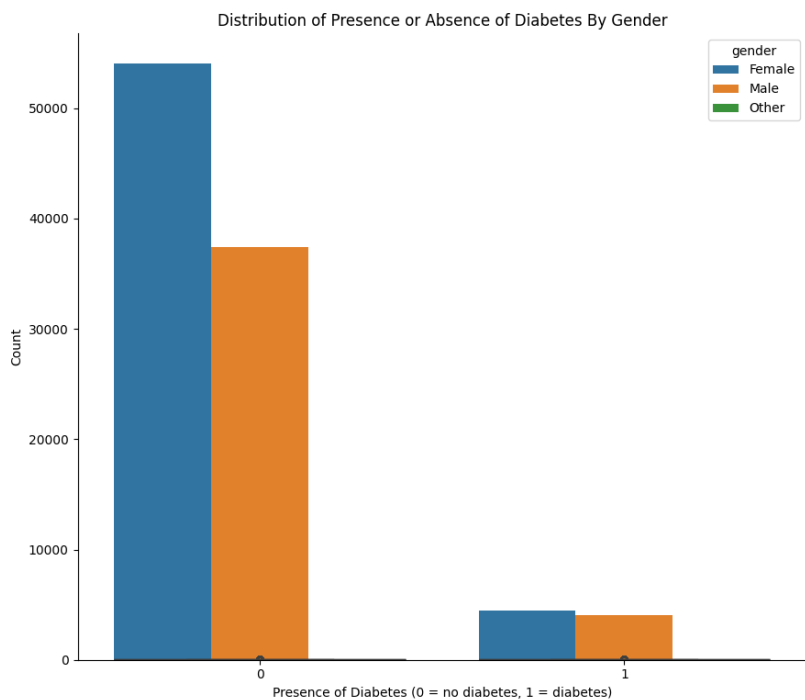


Figure 9 – Data Types of Variables

After Feature Engineering

age	float64
bmi	float64
diabetes	int64
age_range_0-10	int64
age_range_11-20	int64
age_range_21-30	int64
age_range_31-40	int64
age_range_41-50	int64
age_range_51-60	int64
age_range_61-70	int64
age_range_71-80	int64
bmi_range_10-25	int64
bmi_range_26-40	int64
bmi_range_41-55	int64
bmi_range_56-70	int64
bmi_range_71-85	int64
bmi_range_86-100	int64
gender_Female	int64
gender_Male	int64

Figure 10 – Association Rule Mining Table

	antecedents	consequents	support	confidence
0	(bmi_range_41-55)	(diabetes)	0.010832	0.252801
1	(gender_Male, age_range_71-80)	(diabetes)	0.011452	0.226151
2	(gender_Male, age_range_61-70)	(diabetes)	0.010852	0.224964
3	(age_range_71-80, bmi_range_26-40)	(diabetes)	0.019484	0.212224
4	(age_range_71-80)	(diabetes)	0.024904	0.201522
5	(bmi_range_26-40, age_range_61-70)	(diabetes)	0.017543	0.198641
6	(age_range_71-80, bmi_range_26-40, gender_Female)	(diabetes)	0.010282	0.198532
7	(age_range_61-70)	(diabetes)	0.022294	0.197676
8	(age_range_71-80, gender_Female)	(diabetes)	0.013452	0.184423
9	(gender_Female, age_range_61-70)	(diabetes)	0.011442	0.177282
10	(age_range_51-60)	(diabetes)	0.019313	0.130227
11	(bmi_range_26-40, age_range_51-60)	(diabetes)	0.014403	0.127074
12	(gender_Male, bmi_range_26-40)	(diabetes)	0.031926	0.112122
13	(bmi_range_26-40)	(diabetes)	0.063821	0.098805
14	(gender_Male)	(diabetes)	0.040397	0.097490
15	(bmi_range_26-40, gender_Female)	(diabetes)	0.031896	0.088306
16	(gender_Female)	(diabetes)	0.044618	0.076189
17	(age_range_41-50)	(diabetes)	0.010692	0.072910

Figure 11 – Model Accuracy Scores and Confusion Matrices

```
Gender Model Accuracy is 91.62374356153423%
[[ 0 1675]
 [ 0 18322]]
Age Model Accuracy is 91.44871730759614%
[[ 0 1710]
 [ 0 18287]]
BMI Model Accuracy is 91.04365654848228%
[[ 26 1745]
 [ 46 18180]]
Full Model Accuracy is 91.74876231434716%
[[ 50 1598]
 [ 52 18297]]
```

Figure 12 – Gender Model ROC Curve

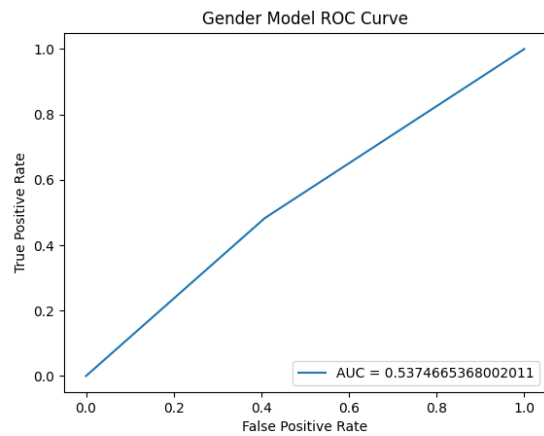


Figure 13 – Age Model ROC Curve

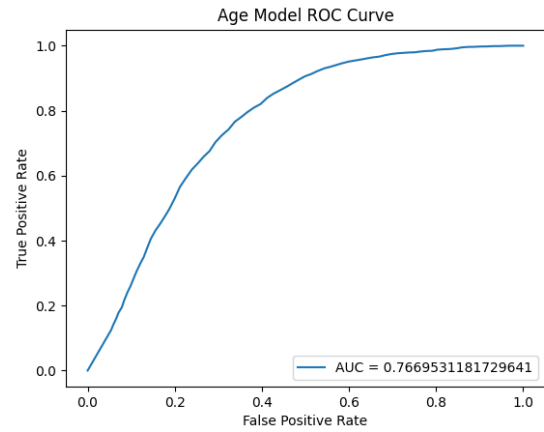


Figure 14 – BMI Model ROC Curve

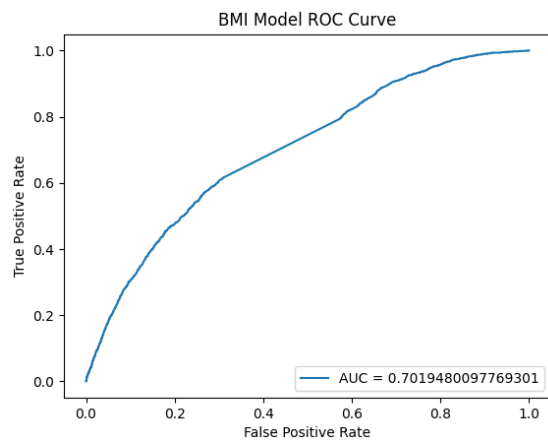


Figure 15 – Full Model ROC Curve

